

Problem Set 1 - Solutions

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Exercise 1

```
library(ellipse)
usair <- read.table("data/usair.txt", head = T, sep = " ")

head(usair)
```

```
##           S02 Neg.Temp Manuf Pop Wind Precip Days
## Phoenix      10   -70.3   213 582  6.0   7.05   36
## Little Rock  13   -61.0    91 132  8.2  48.52  100
## San Francisco 12   -56.7   453 716  8.7  20.66   67
## Denver       17   -51.9   454 515  9.0  12.95   86
## Hartford     56   -49.1   412 158  9.0  43.37  127
## Wilmington   36   -54.0    80  80  9.0  40.25  114
```

```
usair <- usair[, 2:7] #drop the first column
usair[, 1] <- -usair[, 1] #change temp. sign to positive
names(usair)[1] <- 'Temp' #change name of first variable
```

Question 1.1

```
R <- cor(usair); R
```

```
##           Temp           Manuf           Pop           Wind           Precip           Days
## Temp      1.00000000 -0.19004216 -0.06267813 -0.34973963  0.38625342 -0.43024212
## Manuf     -0.19004216  1.00000000  0.95526935  0.23794683 -0.03241688  0.13182930
## Pop       -0.06267813  0.95526935  1.00000000  0.21264375 -0.02611873  0.04208319
## Wind      -0.34973963  0.23794683  0.21264375  1.00000000 -0.01299438  0.16410559
## Precip    0.38625342 -0.03241688 -0.02611873 -0.01299438  1.00000000  0.49609671
## Days     -0.43024212  0.13182930  0.04208319  0.16410559  0.49609671  1.00000000
```

There is a very high positive correlation (~ 0.95) between the “Pop” and “Manuf” variables, suggesting that cities with higher population have more manufacturing enterprises employing at least 20 workers. There are also some others to notice, like the correlation between the average annual temperature (“Temp”) and the “Wind” speed (~ 0.35), the avg. annual precipitations inches (“Precip”) (~ -0.38) and the avg. number of days with precipitations in one year (~ 0.43). Notice also the positive correlation between Temperatures and Precipitation, indicating that the average temperatures decrease when it rains more.

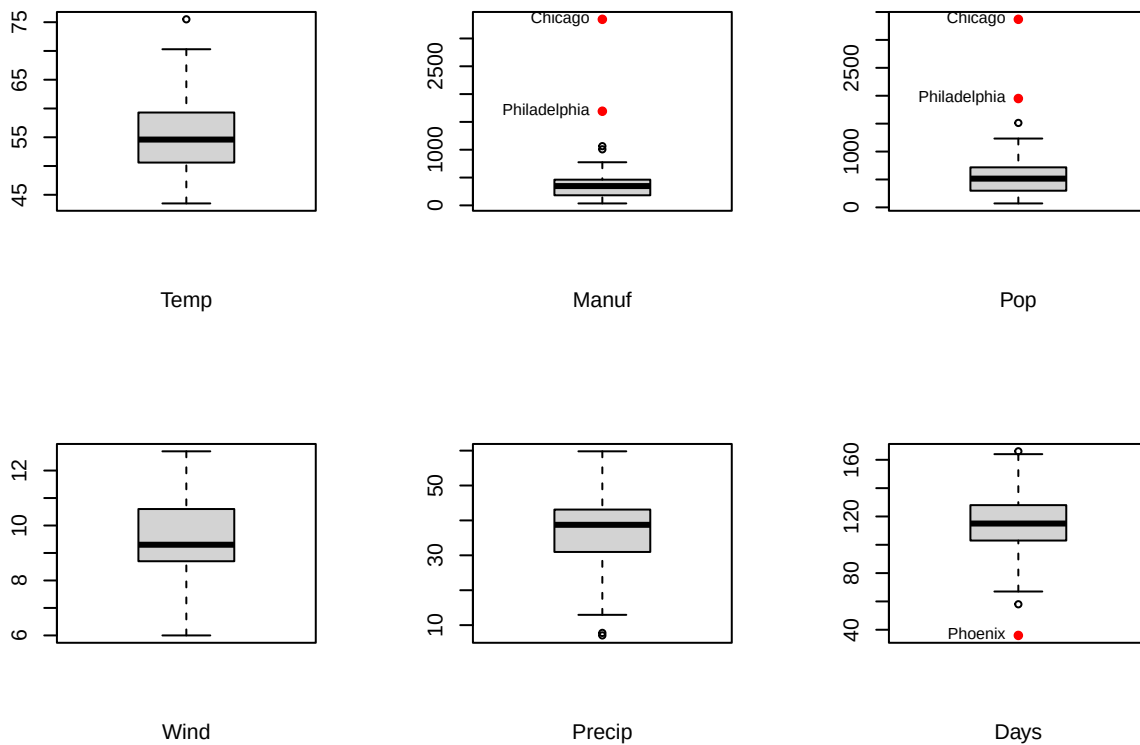
Question 1.2

To select at most 2 outliers per variable, we will change the constant 'const' from the default 1.5 to 2.5; in this way, the outliers selection is stricter.

```
par(mfrow = c(2, 3))
i = 0
const = 2.5

for (i in 1:ncol(usair)) {
  boxplot.data <- boxplot(usair[, i], plot = F)
  Q1 <- boxplot.data$stats[2]
  Q3 <- boxplot.data$stats[4]
  IQR <- Q3 - Q1
  index <- which(usair[, i] < (Q1 - const * IQR) | usair[, i] > (Q3 + const * IQR))
  col.ind <- rep("black", nrow(usair))
  col.ind[index] <- "red"
  out.name <- rownames(usair[index, ])
  boxplot(usair[, i], xlab = names(usair)[i])
  points(rep(1, length(index)), usair[index, i], pch = 16, col = col.ind[index])
  if (length(out.name) > 0) {
    text(x = rep(1, length(index)), y = usair[index, i], out.name, cex = .75, pos = 2)
  }
}
title("Boxplot for each variable", outer = T, line = -2)
```

Boxplot for each variable

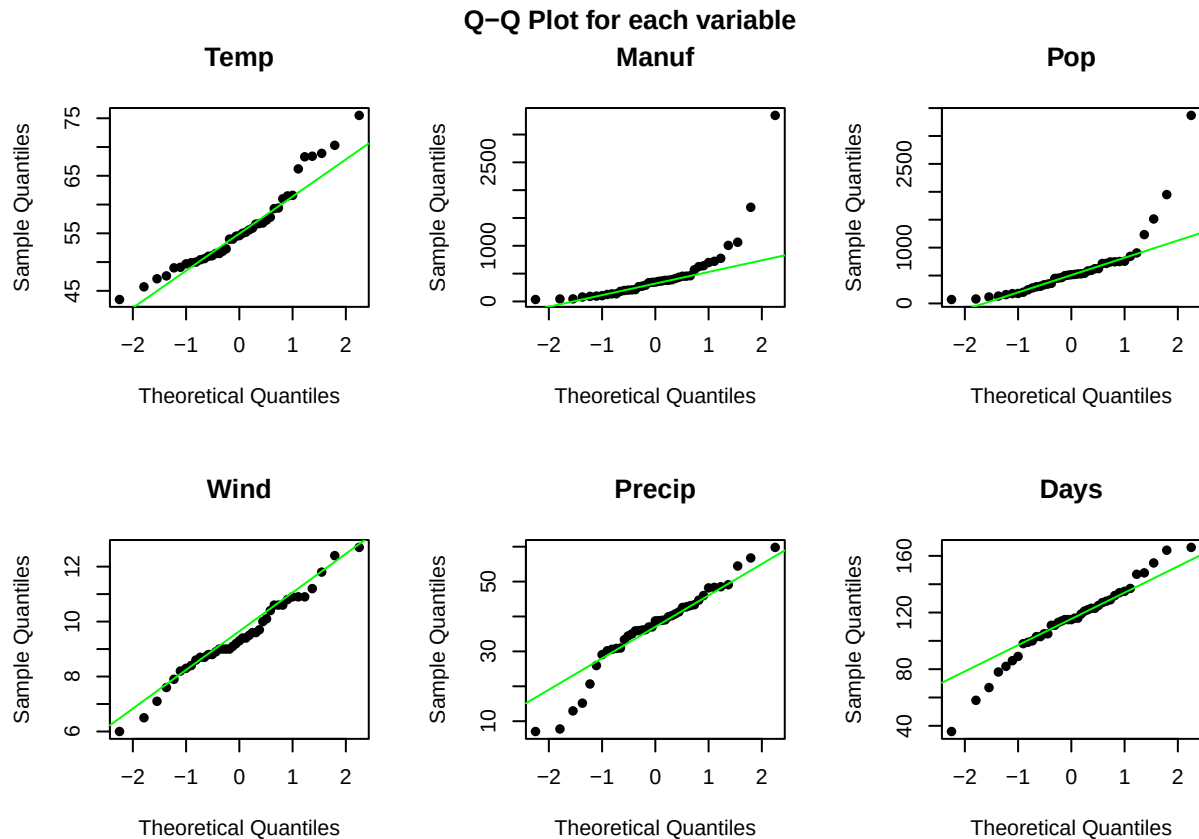


From the above plots, it is clear the presence of some outliers for some variables; we marked in red the non-standard outliers that we will take into consideration from now on, while we left the ones with the white

dot which are the standard ones (const variable equal to 1.5).

Question 1.3

```
par(mfrow = c(2, 3))
for (i in 1:ncol(usair)) {
  qqnorm(usair[, i], pch = 16, main = names(usair)[i])
  qqline(usair[, i], col = "green")
}
title("Q-Q Plot for each variable", outer = T, line = -1)
```



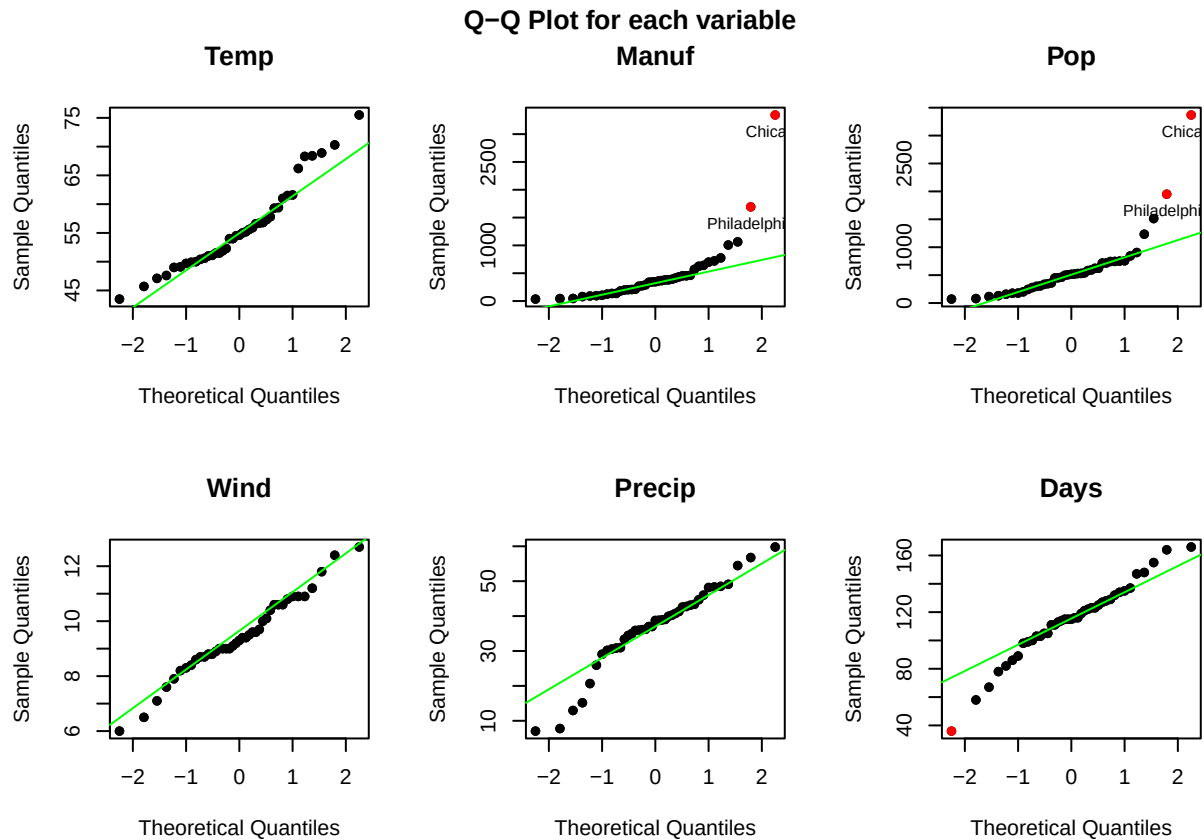
From the Q-Q plots, we see that the variables seem to have a quite normal distribution, with the exception of some outliers, especially in the “Precip”, “Manuf” and “Pop” variables. We can check if these corresponds with the boxplot-identified outliers:

```
par(mfrow = c(2, 3))
for (i in 1:ncol(usair)) {
  boxplot.data <- boxplot(usair[, i], plot = F)
  Q1 <- boxplot.data$stats[2]
  Q3 <- boxplot.data$stats[4]
  IQR <- Q3 - Q1
  index <- which(usair[, i] < (Q1 - const * IQR) | usair[, i] > (Q3 + const * IQR))
  col.ind <- rep("black", nrow(usair))
  col.ind[index] <- "red"
  out.name <- rownames(usair[index, ])
  q <- qqnorm(usair[, i], pch = 16, main = names(usair)[i])
  points(q$x, usair[, i], pch = 16, col = col.ind)
```

```

if (length(out.name) > 0) {
  text(q$x[index], usair[index, i], out.name, cex = .75, pos = 1)
}
qqline(usair[, i], col = "green")
}
title("Q-Q Plot for each variable", outer = T, line = -1)

```



Indeed they do: the marginal distributions appear quite normal, with the exception of some points, which mainly correspond to the outliers found above.

Question 1.4

```

#function to plot correlation for each scatterplot:
panel.cor <- function(x, y, digits = 2, prefix = "", cex.cor, ...)
{
  usr <- par("usr"); on.exit(par(usr))
  par(usr = c(0, 1, 0, 1))
  r <- cor(x, y)
  txt <- format(c(r, 0.123456789), digits = digits)[1]
  txt <- paste0(prefix, txt)
  if(missing(cex.cor)) cex.cor <- 0.8/strwidth(txt)
  text(0.5, 0.5, txt, cex = cex.cor * max(abs(r), 0.2))
}

#creating a vector containing all the univariate outliers
univ.out <- NULL
for (i in 1:ncol(usair)) {

```

```

boxplot.data <- boxplot(usair[, i], plot = F)
Q1 <- boxplot.data$stats[2]
Q3 <- boxplot.data$stats[4]
IQR <- Q3 - Q1
index <- which(usair[, i] < (Q1 - const * IQR) | usair[, i] > (Q3 + const * IQR))
univ.out <- sort(union(index, univ.out))

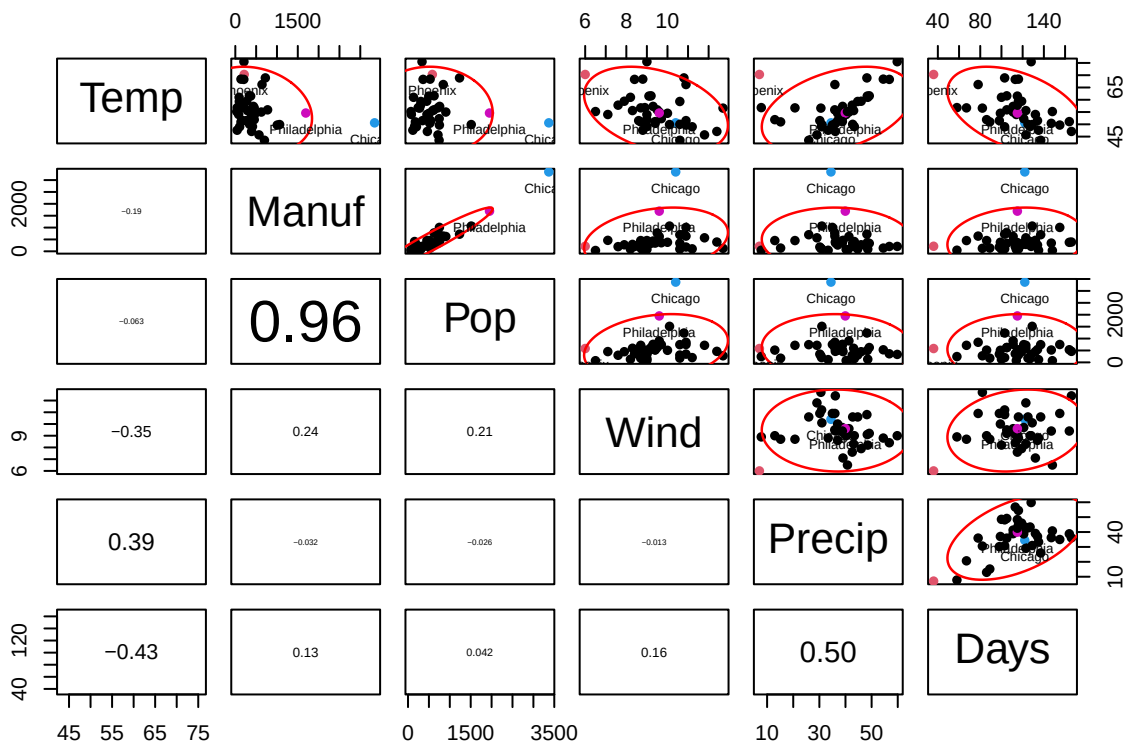
#we have the following univariate outliers:
univ.out

## [1] 1 11 29

col.ind <- rep("black", nrow(usair))
col.ind[univ.out] <- univ.out + 1 #we add 1 because the first color (code 1) is white:

#function used to plot the ellipse in each scatterplot
panel.ellipse <- function(x, y) {
  points(x, y, col = col.ind, pch = 16)
  lines(ellipse(x = cor(x, y), scale = c(sd(x), sd(y)), centre = c(mean(x), mean(y))),
        col = "red", lwd = 1.25)
  text(x[univ.out], y[univ.out], rownames(usair[univ.out, ]), cex = .65, pos = 1)
}
pairs(usair, upper.panel = panel.ellipse, lower.panel = panel.cor)

```



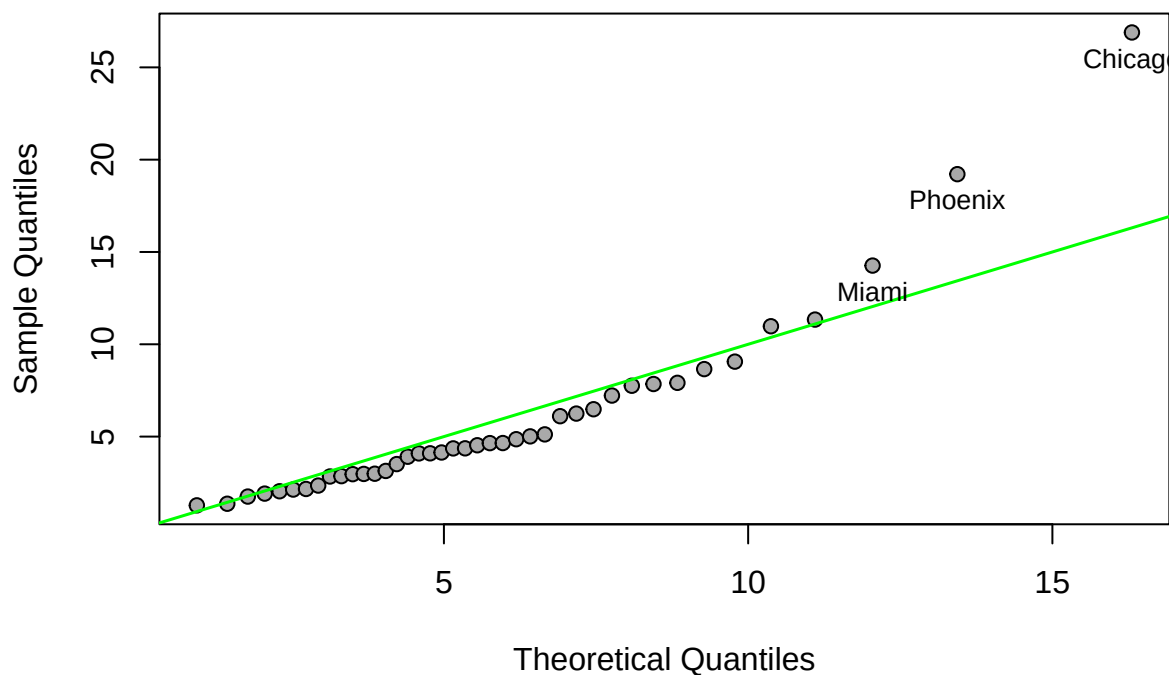
We notice that many times the univariate outliers are the ones which often lie outside the ellipses. However, this does not mean that the points outside the ellipses are bivariate outliers; to spot them, we should use other techniques, such as the chi-squared Q-Q plot or the T² chart and look for unusual observations.

Question 1.5

```
means <- apply(usair, 2, mean)
S <- cov(usair)
d <- mahalanobis(usair, means, S)

chi.q <- qchisq(ppoints(d), df = 6)
par(mfrow = c(1,1))
plot(x = qchisq(ppoints(d), df = 6), y = sort(d), pch = 21, col = "black",
     bg = "darkgrey", main="Chisq Q-Q plot of Mahalanobis distance",
     xlab="Theoretical Quantiles", ylab="Sample Quantiles")
abline(0, 1, col = "green", lwd = 1.5)
df.sorted <- usair[names(sort(d)), ] #dataframe sorted by increasing mahalanobis distances
text(x = qchisq(ppoints(d), df = 6)[39:41], y = sort(d)[39:41],
     rownames(df.sorted[39:41, ]), pos = 1, cex = .8)
```

Chisq Q-Q plot of Mahalanobis distance



We can see that the distribution seems normal, except for 1 or 2 points for which the mahalanobis squared distance is too high, which could signal the presence of some outliers; we marked them on the plot with their names.

Question 1.6

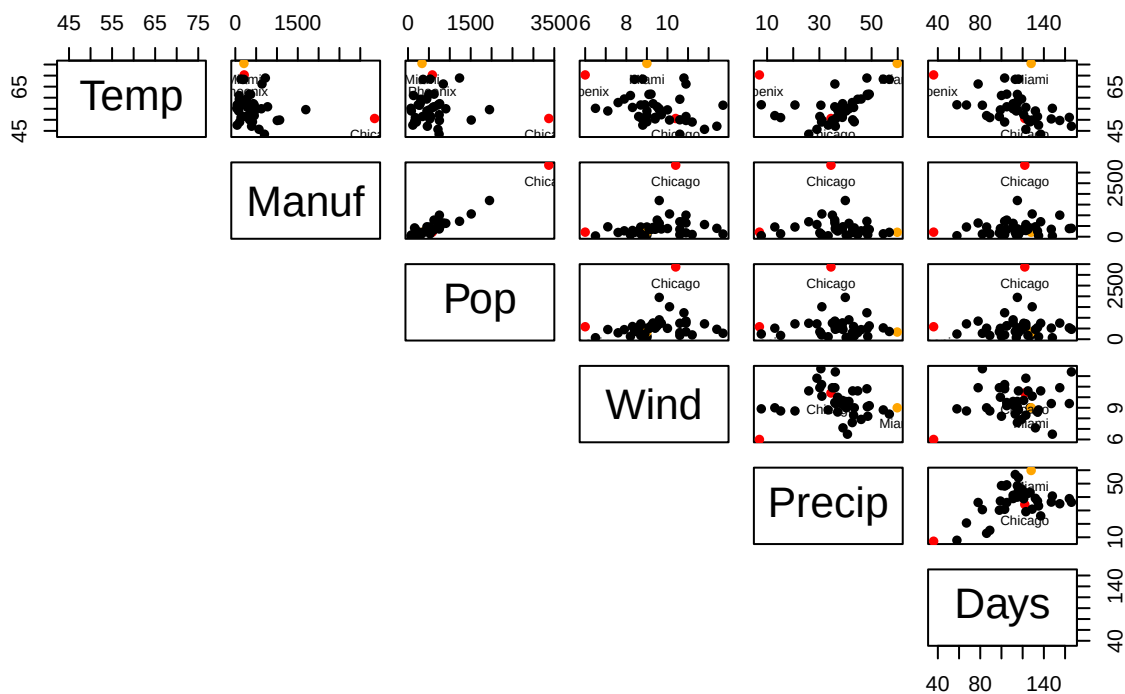
```
panel.text <- function(x, y) {
  points(x, y, col = col.ind2, pch = 16)
  text(x[index.tot], y[index.tot], rownames(usair[index.tot, ]), cex = .65, pos = 1)
}
```

```

index2 <- which(d > qchisq(1-0.01, df = 6))
index2.or <- which(d > qchisq(1-0.05, df = 6))
index2.or <- index2.or[2] #selecting the wanted one (the others are included in index2)
index.tot <- sort(union(index2, index2.or))
col.ind2 <- rep("black", nrow(usair))
col.ind2[index2] <- "red"
col.ind2[index2.or] <- "orange"
pairs(usair, lower.panel = NULL,
      main = "Multivariate outliers in the scatterplot matrix",
      upper.panel = panel.text)

```

Multivariate outliers in the scatterplot matrix

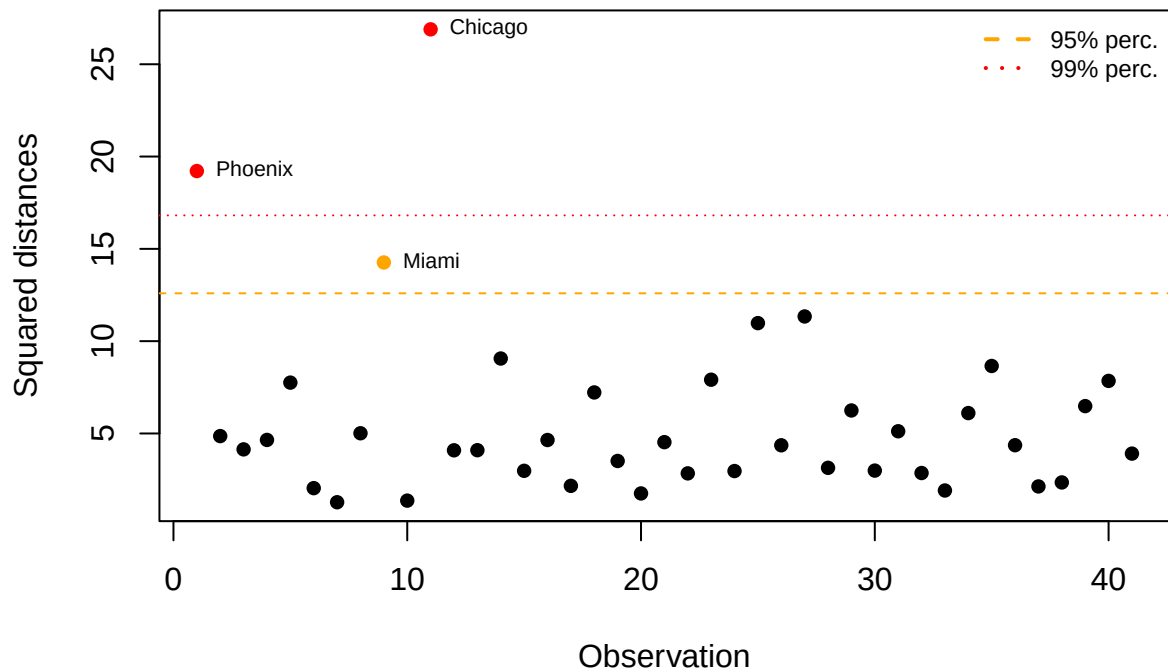


```

plot(d, col = col.ind2, pch = 16, main = "T2-chart", xlab = "Observation",
     ylab = "Squared distances")
abline(h = qchisq(1-0.05, df = 6), lty = 2, col = "orange")
abline(h = qchisq(1-0.01, df = 6), lty = 3, col = "red")
legend("topright", legend = c("95% perc.", "99% perc."), col = c("orange", "red"),
      lwd = 2, lty = c(2, 3), bty = "n", cex = .75)
text(x = seq(1:41)[index.tot], y = d[index.tot], rownames(usair[index.tot, ]),
     pos = 4, cex = .7)

```

T2-chart



As we already found out in point .5, the multivariate outliers seem to be Chicago, Phoenix and Miami. Comparing these with the ones of point 1. (the univariate outliers), we notice that Phoenix and Chicago are both univariate and multivariate outliers, while Miami is only a multivariate outlier and Philadelphia is only a univariate outlier.

Exercise 2

Question 2.1

First of all, we need to find the covariance matrix of X , call it $\mathbf{S}$. We have:

$$Z \sim \mathcal{N}_q(\mathbf{0}, \sigma_z^2 I_q) \implies VZ \sim \mathcal{N}_q(\mathbf{0}, \sigma_z^2 VV^T)$$

Then, since Z and ϵ are independent normal by assumption, we have that:

$$\mathbf{S} = \sigma_z^2 VV^T + \sigma^2 I_p$$

where $\sigma^2 I_p$ is the covariance matrix of $\epsilon \sim \mathcal{N}_p(\mathbf{0}, \sigma^2 I_p)$. Now we focus on finding the eigenvalues of $\mathbf{S}$ (i.e., the variances of the PCs) in order to be able to impose our condition. Notice that we can decompose $\mathbf{S}$ using the spectral decomposition which gives us that $\mathbf{S} = P\Lambda P^T$, which in turn leads to the following diagonalization:

$$\Lambda = P^T \mathbf{S} P = \text{diag}(\lambda_1, \dots, \lambda_p),$$

where Λ are the eigenvalues of $\mathbf{S}$ and P is the loadings matrix. Now recall that the total variance explained is the total variance of the PCs, given by:

$$\text{trace}(\mathbf{S}) = \lambda_1 + \dots + \lambda_p$$

which in our case are given by:

$$\text{trace}(\mathbf{S}) = (p(\sigma_z^2 + \sigma^2) + (p - q)\sigma^2).$$

The condition we have to impose is the following:

$$\frac{\sum_{i=1}^q \lambda_i}{\sum_{i=1}^p \lambda_i} > 0.8$$

which can be written as follows:

$$\sum_{i=1}^q \lambda_i > 0.8(\sum_{i=1}^q \lambda_i + \sum_{j=q+1}^p \lambda_j).$$

Finally, substituting we get:

$$q((1 + \delta)\sigma^2 + \sigma^2) > 0.8(q((1 + \delta)\sigma^2 + \sigma^2) + (p - q)\sigma^2).$$

Solving this inequality for δ yields the following result:

$$\delta > 4\frac{p}{q} - 6,$$

and, recalling that $\delta > -1$ by assumption, we get that $\frac{p}{q} > \frac{5}{4}$.

Question 2.2

We have that $Z = (Z_1, Z_2) \sim \mathcal{N}_2(\mu_z, \Sigma_z)$, where:

$$\mu_z = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_z = \sigma_z^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

We want to find the distribution of VZ first, and then the one of X , in order to be able to compute the conditional distribution of interest. We will have that $VZ \sim \mathcal{N}_2(\mu_{VZ}, \Sigma_{VZ})$, where:

$$\mu_{VZ} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_{VZ} = VI_2V^T = \begin{bmatrix} \frac{5}{9} & -\frac{4}{9} & \frac{2}{9} \\ -\frac{4}{9} & \frac{5}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{2}{9} & \frac{5}{9} \end{bmatrix}$$

We will now consider ϵ which is such that: $\epsilon \sim \mathcal{N}_3(\mathbf{0}, \Sigma_\epsilon) = \mathcal{N}_3(\mathbf{0}, \frac{1}{3}I_3)$, where I_3 is the 3×3 identity matrix. Notice that Z was of dimension $2 = q < p = 3$ by the exercise assumption. Now we can finally find the multivariate normal distribution of X : since X was given by the transformation $X = \mu + VZ + \epsilon$, we get that: $X = (X_1, X_2, X_3) \sim \mathcal{N}_3(\mu_{VZ\epsilon}, \Sigma_{VZ\epsilon})$, where:

$$\mu_{VZ\epsilon} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \Sigma_{VZ\epsilon} = \Sigma_{VZ} + \Sigma_{\epsilon} = \begin{bmatrix} \frac{8}{9} & -\frac{4}{9} & \frac{2}{9} \\ -\frac{4}{9} & \frac{8}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{2}{9} & \frac{11}{9} \end{bmatrix}$$

Now, to find the distribution $(X_1, X_2 | X_3 = -1)$, we recall the conditional distribution will be distributed as follows:

$$X_1 | X_2 = x_2 \sim \mathcal{N}(\mu_{1|2}, \Sigma_{1|2})$$

where:

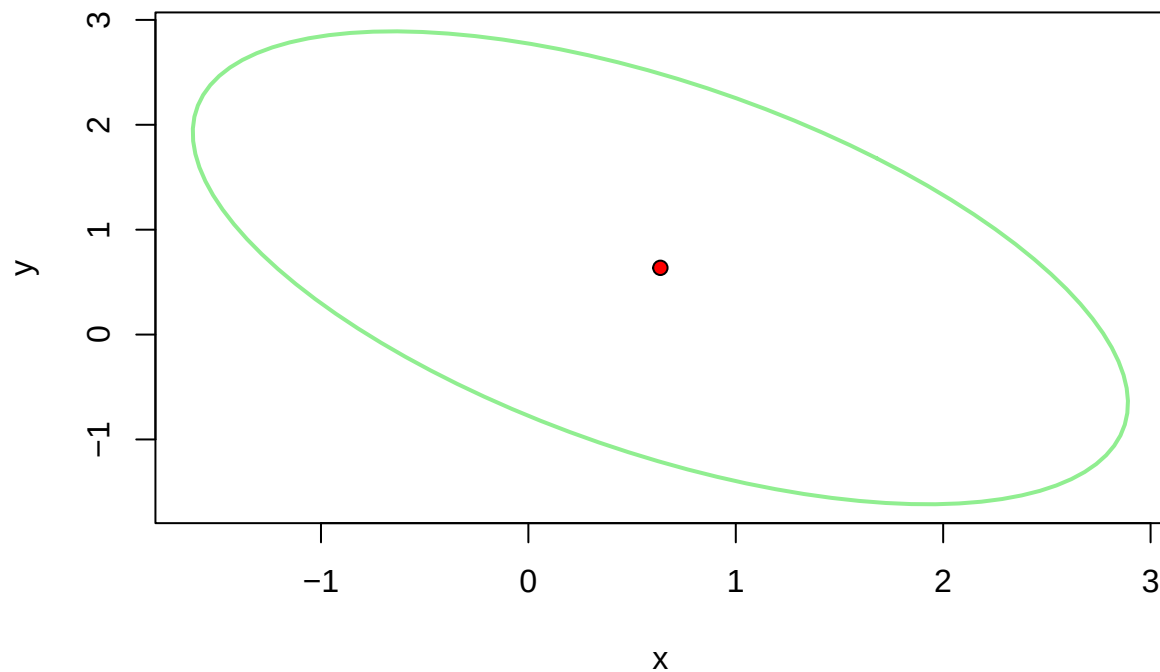
$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Thus we compute the parameters using R:

```
#Computation:
library(ellipse)
mu <- c(1, 1, 1)
V <- 1/3 * matrix(c(-1, 2, 2, -1, 2, 2), ncol = 2, nrow = 3, byrow = T)
I2 <- matrix(c(1, 0, 0, 1), nrow = 2, ncol = 2)
SigmaVZ <- V%*%I2%*%t(V)
I3 <- 1/3 * matrix(c(1, 0, 0, 0, 1, 0, 0, 0, 1), nrow = 3, ncol = 3)
SigmaVZeps <- SigmaVZ + I3
mu1 <- mu[1:2]
mu2 <- mu[3]
Sigma11 <- SigmaVZeps[1:2, 1:2]
Sigma12 <- SigmaVZeps[1:2, 3]
Sigma21 <- SigmaVZeps[3, 1:2]
Sigma22 <- SigmaVZeps[3, 3]

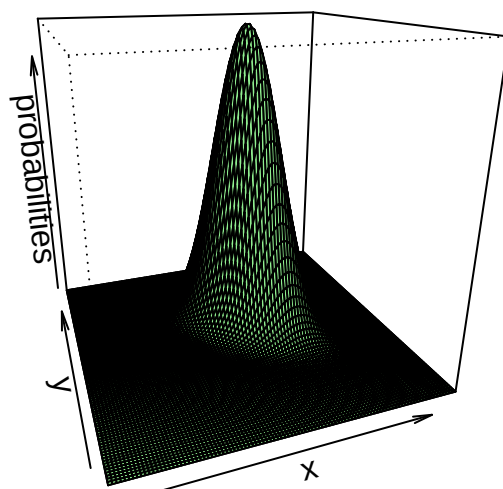
#Conditional distribution parameters:
mu_1given2 = mu1 + Sigma12%*%solve(Sigma22)%*%(-1 - mu2)
Sigma_1give2 = Sigma11 - Sigma12%*%solve(Sigma22)%*%Sigma21

#Plotting:
plot(ellipse(x = Sigma_1give2, centre = mu_1given2), type = 'l', lwd = 2,
      col = 'lightgreen')
points(mu_1given2[1, ], mu_1given2[2, ], pch = 21, bg = 'red')
```



```
mu_1given2 <- rep(mu_1given2[1], 2) #we do so to be able to use dmmnorm()
library(mnormt)
x <- seq(-5, 5, by = .1)
y <- seq(-5, 5, by = .1)
f <- function(x, y) dmmnorm(x = cbind(x, y), mean = mu_1given2, varcov = Sigma_1give2)
probabilities <- outer(x, y, f)
persp(x, y, probabilities, col = 'lightgreen', theta = -20, phi = 20,
      border = 'black', main = 'Conditional distribution plot')
```

Conditional distribution plot



Exercise 3

```
rm(list = ls())
pendigits <- read.table("data/pendigits.txt", head = F, sep = ",")
names(pendigits) <- c(paste0(rep(c("x", "y"), 8), rep(1:8, each=2)), "digit")
head(pendigits)
```

```
##      x1  y1 x2  y2 x3  y3 x4  y4 x5  y5 x6  y6 x7  y7 x8  y8 digit
## 1  47 100 27  81  57  37  26   0  0 23  56 53 100 90  40 98     8
## 2   0  89 27 100  42  75  29  45 15 15  37  0  69  2 100  6     2
## 3   0  57 31  68  72  90 100 100 76 75  50 51  28 25  16  0     1
## 4   0 100  7  92   5  68  19  45 86 34 100 45  74 23  67  0     4
## 5   0  67 49  83 100 100  81  80 60 60  40 40  33 20  47  0     1
## 6 100 100 88  99  49  74  17  47  0 16  37  0  73 16  20 20     6
```

Question 3.1

```
pendigits.pca <- prcomp(pendigits[1:16], scale = T)
pendigits.pca$sdev
```

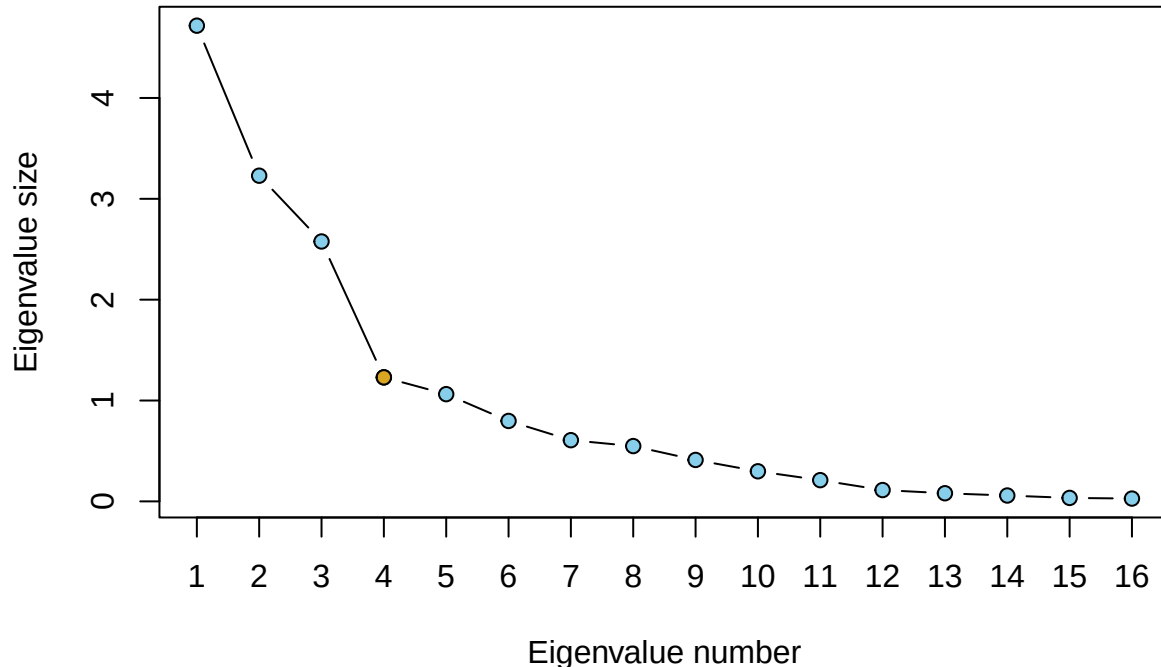
```
## [1] 2.1717703 1.7969730 1.6052434 1.1089372 1.0310671 0.8929386 0.7788784
## [8] 0.7404401 0.6409556 0.5461052 0.4588166 0.3351073 0.2836918 0.2408363
## [15] 0.1851067 0.1667316
```

```
summary(pendigits.pca)
```

```
## Importance of components:
##               PC1    PC2    PC3    PC4    PC5    PC6    PC7
## Standard deviation  2.1718 1.7970 1.6052 1.10894 1.03107 0.89294 0.77888
## Proportion of Variance 0.2948 0.2018 0.1610 0.07686 0.06644 0.04983 0.03792
## Cumulative Proportion 0.2948 0.4966 0.6577 0.73452 0.80096 0.85079 0.88871
##               PC8    PC9    PC10    PC11    PC12    PC13    PC14
## Standard deviation  0.74044 0.64096 0.54611 0.45882 0.33511 0.28369 0.24084
## Proportion of Variance 0.03427 0.02568 0.01864 0.01316 0.00702 0.00503 0.00363
## Cumulative Proportion 0.92297 0.94865 0.96729 0.98045 0.98747 0.99250 0.99612
##               PC15    PC16
## Standard deviation  0.18511 0.16673
## Proportion of Variance 0.00214 0.00174
## Cumulative Proportion 0.99826 1.00000
```

```
plot(pendigits.pca$sdev^2, type = "b", xlab = "Eigenvalue number",
     ylab = "Eigenvalue size", main = "Screeplot", axes = F, pch = 21, bg = "skyblue")
points(4, pendigits.pca$sdev[4]^2, pch = 21, bg = "goldenrod")
box()
axis(1, at = 1:16)
axis(2, tick = T)
```

Screeplot



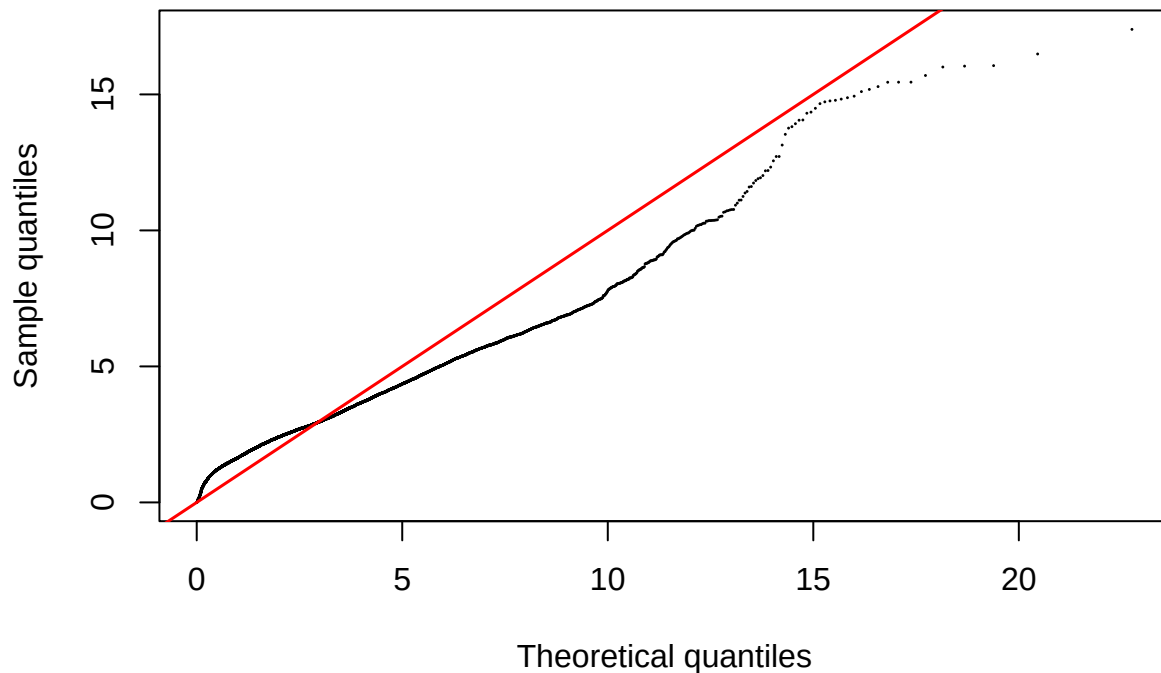
We need to select PCs in order to cover at least 70% of the total variance, so at least we need to select the first $k = 4$ PCs; however, notice that adding only two more PCs we get over 80%, which is again a good compromise since the original variables were $p = 16$, so there is a good variable reduction anyway. We have chosen to consider only the first $k = 4$ PCs.

Question 3.2

If the normality assumption is satisfied, we should be able to check that also using the principal components, since they are linear transformations of the original variables. The following are the chi-squared Q-Q plots of the first three PCs:

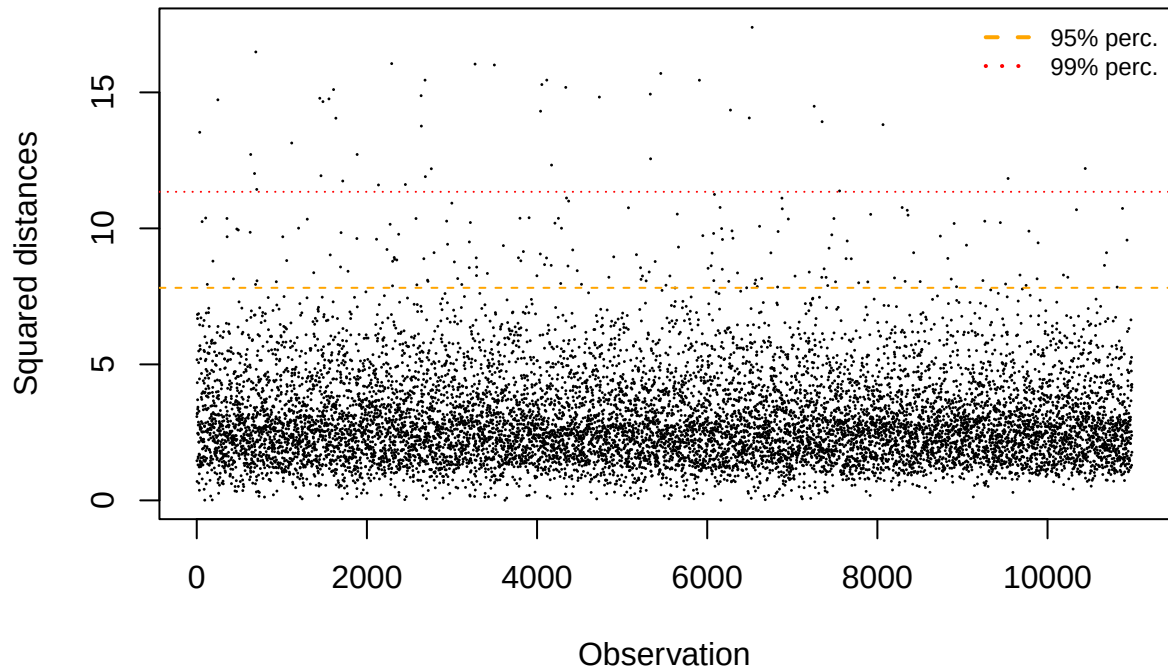
```
X <- pendigits.pca$x[, 1:3]
bar.X <- colMeans(X) #theoretically they should be 0, in practice almost 0!
S.X <- diag(pendigits.pca$sdev[1:3]^2)
d.X <- mahalanobis(X, bar.X, S.X)
plot(x = qchisq(ppoints(d.X), df = 3), y = sort(d.X),
     main = "Chi-squared Q-Q plot of Mahalanobis distance of the first three PCs",
     xlab = "Theoretical quantiles",
     ylab = "Sample quantiles", pch = 16, cex = .2)
abline(0, 1, col = "red", lwd = 1.5)
```

Chi-squared Q-Q plot of Mahalanobis distance of the first three PC



```
#T2-chart
plot(d.X, pch = 16, main = "T2-chart", xlab = "Observation", ylab = "Squared distances",
     cex = .2)
abline(h = qchisq(1-.05, df = 3), lty = 2, col = "orange")
abline(h = qchisq(1-.01, df = 3), lty = 3, col = "red")
legend("topright", legend = c("95% perc.", "99% perc."), col = c("orange", "red"),
     lty = c(2, 3), lwd = 2, cex = .75, bty = "n")
```

T2-chart



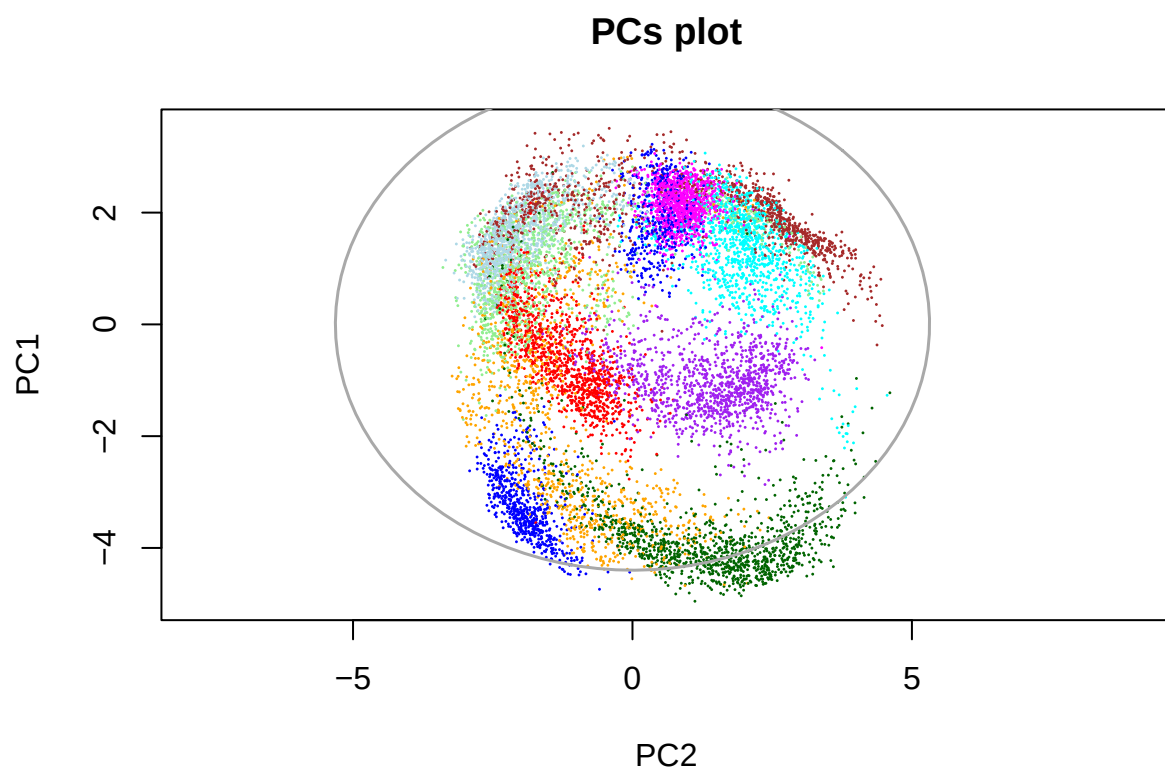
From these two plots, we can see that data are mostly normally distributed, with the exception of some possible outliers, that we'll see later in more detail.

Question 3.3

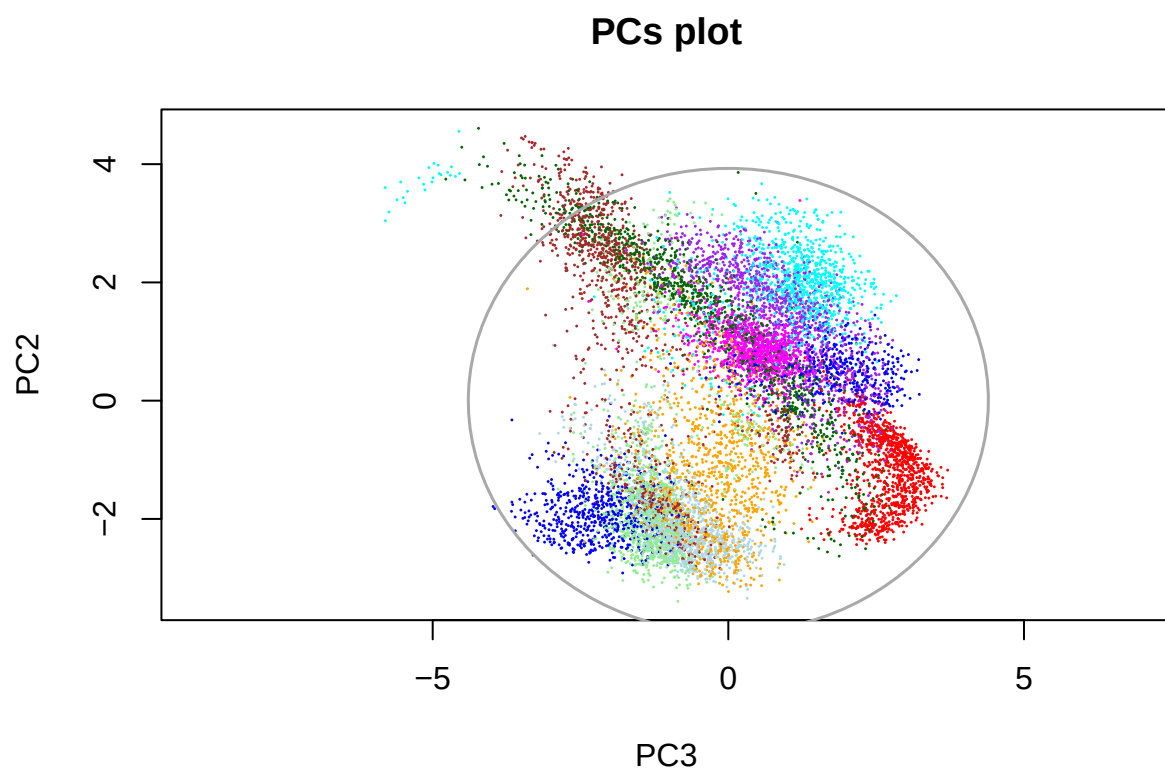
```
#We extract the first three PCs:
PC1 <- pendigits.pca$x[, 1]
PC2 <- pendigits.pca$x[, 2]
PC3 <- pendigits.pca$x[, 3]

col.ind <- rep('black', nrow(pendigits))
lookup<-c("darkgreen", "brown", "lightblue", "magenta", "purple",
          "blue", "red", "lightgreen", "orange", "cyan")
for (i in 0:9) {
  col.ind[pendigits$digit == i] <- lookup[i + 1]
}

#Plotting:
plot(PC1 ~ PC2, main = "PCs plot", pch = 16, cex = .2, asp = 1, col = col.ind)
lines(ellipse(x = diag(pendigits.pca$sdev[1:2])^2, centre = c(0, 0), level = .95),
      lwd = 1.5, col = "darkgrey")
```

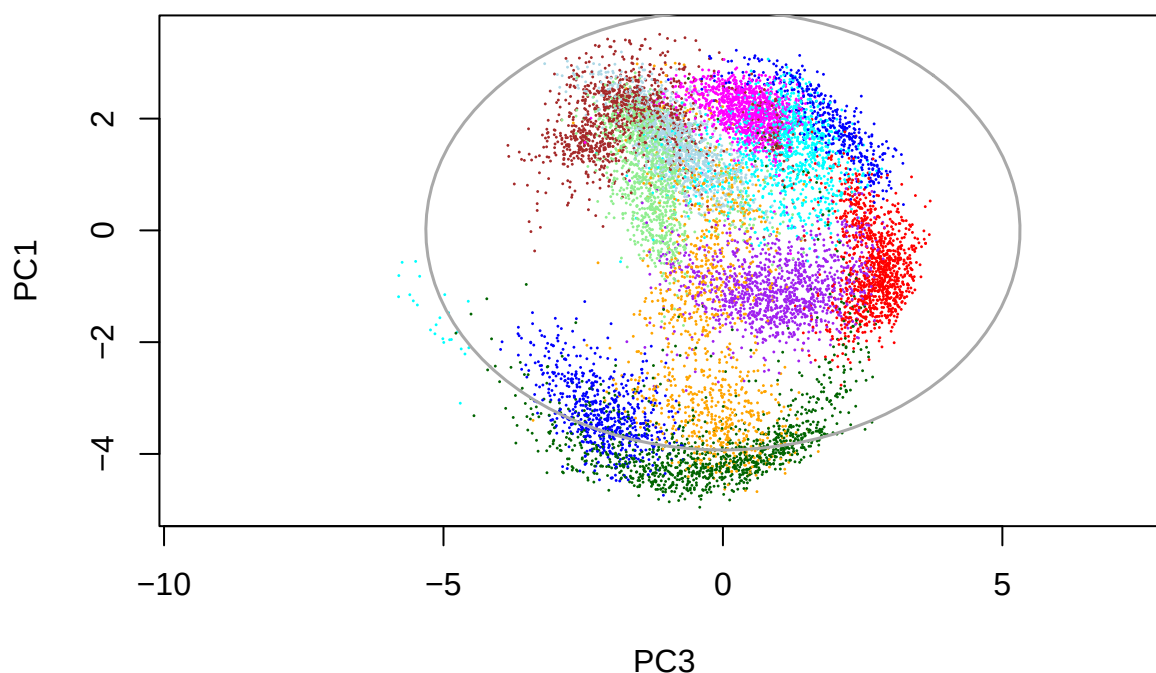


```
plot(PC2 ~ PC3, main = "PCs plot", pch = 16, cex = .2, asp = 1, col = col.ind)
lines(ellipse(x = diag(pendigits.pca$sdev[2:3])^2, centre = c(0, 0), level = .95),
      lwd = 1.5, col = "darkgrey")
```



```
plot(PC1 ~ PC3, main = "PCs plot", pch = 16, cex = .2, asp = 1, col = col.ind)
lines(ellipse(x = diag(pendigits.pca$sdev[c(1, 3)])^2, centre = c(0, 0), level = .95),
      lwd = 1.5, col = "darkgrey")
```

PCs plot

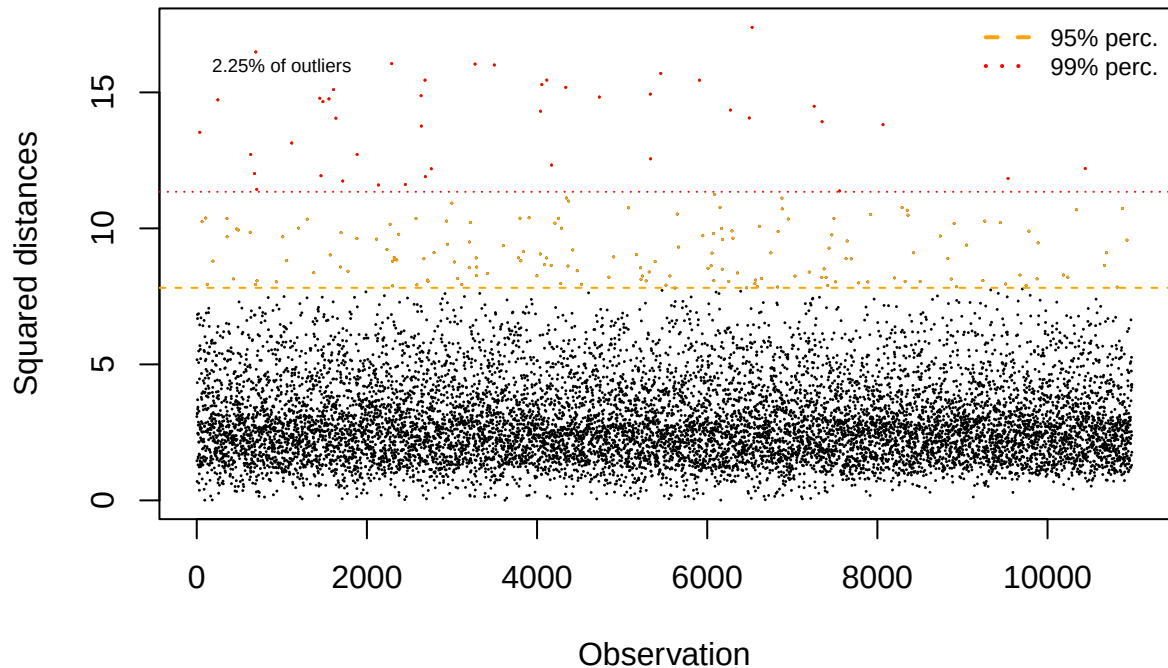


Bivariate normality appears to be respected, except for the dark green observations. They seem to be possible outliers, as seen in the first and last graphs.

Question 3.4

```
#T^2 chart
index.out.red <- which(d.X > qchisq(1-.01, df = 3))
index.out.orange <- which(d.X > qchisq(1-.05, df = 3))
prop <- (length(index.out.red) + length(index.out.orange))/nrow(X) * 100
plot(d.X, pch = 16, main = "T2-chart", xlab = "Observation", ylab = "Squared distances",
     cex = .2)
points(seq(1:nrow(pendigits))[index.out.orange], d.X[index.out.orange],
       col = 'orange', pch = 16, cex = .2)
points(seq(1:nrow(pendigits))[index.out.red], d.X[index.out.red],
       col = 'red', pch = 16, cex = .2)
abline(h = qchisq(1-.05, df = 3), lty = 2, col = "orange")
abline(h = qchisq(1-.01, df = 3), lty = 3, col = "red")
legend("topright", legend = c("95% perc.", "99% perc."), col = c("orange", "red"),
      lty = c(2, 3), lwd = 2, cex = .75, bty = "n")
text(1000, 16, paste0(round(prop, 2), '%', ' of outliers'), cex = .6)
```

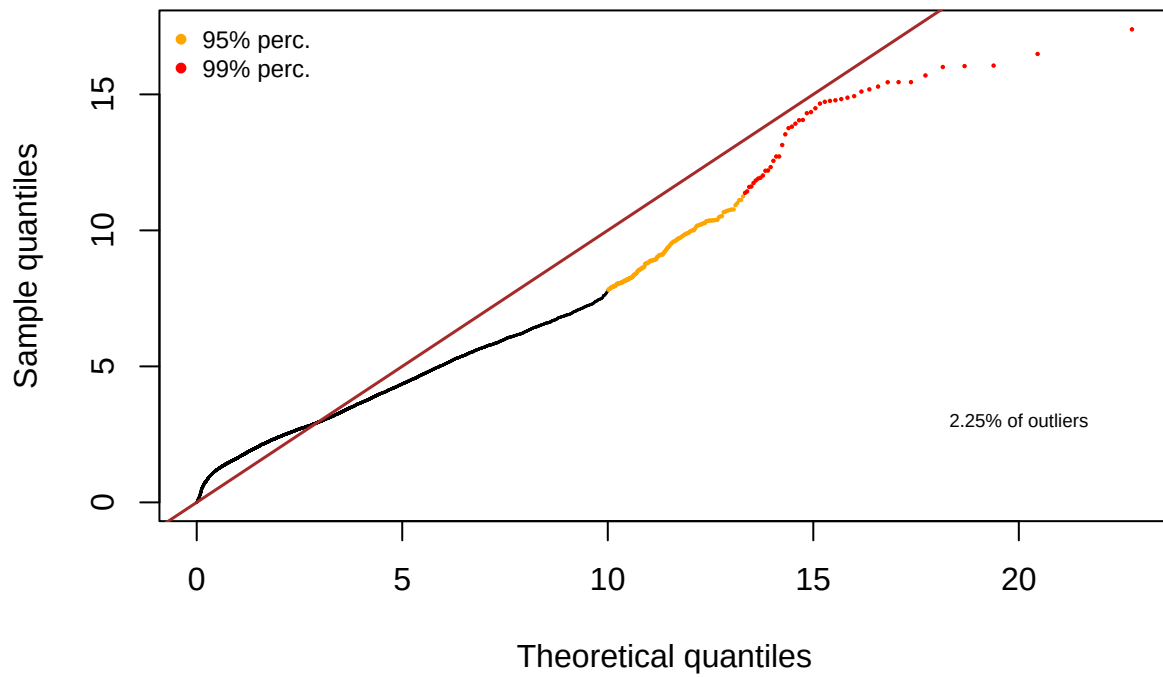
T2-chart



```
#Chi-squared Q-Q plot
index.out.red <- which(sort(d.X) > qchisq(1-.01, df = 3))
index.out.orange <- which(sort(d.X) > qchisq(1-.05, df = 3))

qqplot(qchisq(ppoints(d.X), df = 3), d.X,
       main = "Chi-squared Q-Q plot of Mahalanobis distance of the first three PCs",
       xlab = "Theoretical quantiles",
       ylab = "Sample quantiles", pch = 16, cex = .2)
points(qchisq(ppoints(d.X), df = 3)[index.out.orange], sort(d.X)[index.out.orange],
       col = 'orange', pch = 16, cex = .3)
points(qchisq(ppoints(d.X), df = 3)[index.out.red], sort(d.X)[index.out.red],
       col = 'red', pch = 16, cex = .3)
abline(0, 1, col = "brown", lwd = 1.5)
legend("topleft", legend = c("95% perc.", "99% perc."), col = c("orange", "red"),
       pch = c(16, 16), cex = .75, bty = "n")
text(20, 3, paste0(round(prop, 2), '%', ' of outliers'), cex = .6)
```

Chi-squared Q-Q plot of Mahalanobis distance of the first three PC



There are some possible outliers, but compared to the number of observations of the entire data frame, they are not so much: indeed only 2.25% of all the observations are possible multivariate outliers.